Endogenous Investment and Capital Accumulation in a Disaggregated Keynesian Cross

A Normalized-CES Agent-Based Model with an Endogenous Labor Market

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Abstract

We extend Teglio (2025)'s disaggregated Keynesian cross with endogenous investment and capital accumulation. Production is a normalized CES (Klump and La Grandville, 2000) with elasticity σ; firms hire endogenously, the wage follows a Blanchflower-Oswald curve, and the model is stock-flow consistent throughout.

Three results: (1) $Y(\rho)$ is U-shaped: its turning point ρ^* rises with σ, and the anchored retention rate lies left of that turn at every elasticity tested, so the marginal effect is negative, the wide-range effect positive. Through a single slope this is a sign frontier $\sigma^* \approx 0.65$, a declared reduction, not a verdict. (2) The region where retention depresses output is conditional on fiscal institutions: a balanced-budget benefit at an OECD-band replacement rate removes it. (3) The wage curve generates a fragility immune to demand instruments: three stabilization hypotheses fail through one wage $\rightarrow$ unemployment $\rightarrow$ capital-erosion channel. That immunity is diagnostic, the fragility is a relative-price phenomenon, not a demand one, and it is bracketed by the price convention: it stands under a fixed numéraire, and is switched off, through a Pigou effect, under complete instantaneous pass-through, with neither convention anchored.

A sixteen-parameter global sensitivity analysis separates what generalizes from what does not. The negative margin at the anchored ρ survives marginalization where the turn resolves. The frontier does not: σ is nearly inert once other parameters move, so a sign crossing there is conditional, not structural. Viability is governed almost entirely by the depreciation rate, at whose empirically implied value no configuration survives.

Keywords: agent-based macroeconomics; stock-flow consistency; elasticity of substitution; wage-led growth; global sensitivity analysis; capital accumulation.

JEL: E11, E12, E22, C63, D31.

*Correspondence: `904394@stud.unive.it`. This paper documents an exploratory computational model. Every number reported here is either measured from a committed, reproducible driver script, or explicitly declared as a design target or convention; the distinction is maintained throughout (Section 5). Code, data and figures: `https://github.com/MattiaSaramin/endogenous-investment-abm`.

Contents

1 Introduction

The disaggregated Keynesian cross of Teglio (2025) shows how heterogeneity in the marginal propensity to consume opens an output gap: when income accrues to agents who do not spend it, demand leaks out of the circular flow and the economy settles below capacity. The mechanism is a demand mechanism, and the supply side plays no active role: capacity is given. That paper's conclusions name endogenous investment out of capitalists' accumulated wealth, affecting both the demand and the supply sides, as future work for the model, and venture a conjecture about its outcome; the present extension is that development, and Section 6 returns to the conjecture.

This paper asks what happens when the supply side is made endogenous in the most direct way available, by letting firms invest out of their own profits and accumulate capital. The extension is not merely additive. Once capital accumulates and labor is hired endogenously, two channels that point in opposite directions become active simultaneously.

On the supply side, higher retention ρ means more investment, more capital, and higher productive capacity; taken alone, this raises output. On the demand side, in a demand-constrained regime, more capital means fewer workers are needed to serve the same demand: employment falls, the wage bill falls, and because workers have a higher propensity to consume than capitalists, demand falls with it.

Which channel dominates is the classical wage-led versus profit-led question (Bhaduri and Marglin, 1990; Lavoie and Stockhammer, 2013). Posed as a question about a sign, it invites an answer in the form of a label. This paper's organizing finding is that the label is the wrong object: what the model determines is the shape of the response of output to retention, and the sign is a local reading of that shape.

Three contributions. First, $Y(\rho)$ is U-shaped, and the turn is where the economics is (Section 6). The response has a turning point ρ^* that lies inside the swept support and rises with σ, from 0.42 to 0.64 across the range examined. The anchored retention rate (derived from the investment-rate anchor in Section 4.2, not assumed) lies to the left of that turn at every elasticity tested: the marginal effect of retention at the point where the data put the economy is negative, while its effect over a wide enough range is positive. Both are properties of one curve. Summarized through a single slope, this structure presents as a sign frontier $\sigma^* \approx 0.65$, a reduction we retain, and declare as one.

Second, the region in which retention depresses output is conditional on fiscal institutions (Section 8). A balanced-budget unemployment benefit, at a replacement rate inside the OECD band, eliminates that region entirely, making retention expansionary at every elasticity tested. The leakage of the unemployed from the circular flow is the engine of the effect, and closing the leak removes it; a sign reported without its institutional setting is therefore incomplete.

Third, the wage curve is a source of fragility immune to demand instruments (Sections 7 and 10). Making wages flexible destabilizes the economy through a wage $\rightarrow U \rightarrow$ capital-erosion feedback, and three separate stabilization hypotheses aimed at it (slower demand expectations, the unemployment benefit, and that benefit as a cushion against firm heterogeneity) are each falsified by the data designed to test them, all three through that same channel. The resistance to demand instruments is diagnostic rather than incidental: the feedback is a relative-price phenomenon, not a demand one. A price probe shows this directly: holding the numéraire fixed is itself a convention, and relaxing it to complete instantaneous pass-through switches the feedback off through a Pigou effect on real balances, so the finding is bracketed by the pass-through convention, with neither end anchored (Section 10.4).

A fourth contribution is methodological and kept deliberately off the main line of argument: the sign depends on the estimator used to measure it, a two-point chord and a whole-support slope disagreeing on 108 identified design points under a verdict rule fixed in advance (Section 9.5). The same analysis shows that the frontier does not survive marginalization, viability is governed almost entirely by the depreciation rate rather than by σ, and at the rate the data imply, no sampled configuration remains viable (Section 9).

A note on what this paper reports. An unusual proportion of what follows is negative: three hypotheses are falsified, one headline does not generalize, and one measurement instrument turns out to carry part of its own answer. This is by design, under a declared rule on what may be reported as a result (Section 5); the claims withdrawn under that rule are listed in Section A.4 rather than quietly dropped.

The remainder is organized as follows. Section 2 places the model in the literature and Section 3 presents it. Section 4 documents parameter anchoring and Section 5 the experimental protocol. Section 6 to Section 9 report the four blocks of results, Section 10 draws the cross-cutting readings, Section 11 states the limitations, and Section 12 concludes. Section A collects the validation record.

2 Relation to the literature

Agent-based and stock-flow-consistent macroeconomics. The model belongs to the tradition of macroeconomic ABMs that impose stock-flow consistency on a disaggregated economy (Godley and Lavoie, 2007; Caiani et al., 2016), and shares with the Schumpeter-meeting-Keynes family (Dosi et al., 2010) the ambition of letting aggregate regularities emerge from interacting heterogeneous agents rather than from a representative optimizer. It is deliberately smaller than those models: there is one good, one price (the numéraire), no credit and no entry or exit. The purpose is not to reproduce a full macro-economy, but to isolate a single mechanism, endogenous capital accumulation inside a demand-driven economy, with enough structure to make its sign an empirical question.

Two of those simplifications are omissions relative to the base model in particular. In Teglio (2025) household rationality (four graded levels, from zero-intelligence to semi-rational) and the symmetry of the interaction network are the independent variables of the study, not implementation details; here both are held fixed. The choice is deliberate: to attribute a measured effect to the accumulation channel the interaction topology has to be held still, or the effect is not attributable to anything. It is not costless, and the cost is measured internally rather than asserted: Section 7.4 shows the heterogeneity cascade running precisely through the fixed spending shares, the very object frozen here, so that with the network held symmetric a failing firm's demand is destroyed rather than rerouted. The fixed price inherited from the base model calls for the same genealogical note: there the numéraire is innocuous, because the wage is not a price but a share of revenues distributed in equal parts to a firm's employees (Teglio, 2025), so "real equals nominal" costs nothing when no relative price moves. A wage curve changes that, once the wage responds to unemployment it is itself a relative price, and holding the numéraire fixed is no longer free, which is why the wage-flexibility result is reported as bracketed by the pass-through convention (Section 10.4) rather than as a property of the labor market alone.

These inherited choices, and this paper's departures from them, are collected in Table 1.

Table 1: The model against its base model, Teglio (2025), dimension by dimension. The comparison is qualitative: every entry is a structural feature, not a measured quantity. The first four rows are the extensions this paper contributes; the next two are simplifications relative to the base model, whose two independent variables are held fixed here (Section 7.4); the last three record further differences in inventories, fiscal design and method.

Dimension	Teglio (2025)	This paper
Capacity	Labor only; output is demand-determined, with no capital	Normalized CES in capital and labor (Eq. (1)), with the elasticity σ swept
Employment	Fixed job links - "a firm has always the same number of employees"	Endogenous: the smaller of a demand, a profit-maximizing and a workforce limit (Eq. (2)), with firing into a pool and random matching
Wage	Not a relative price; a firm's revenues are split among its workers in equal parts	Endogenous, set by a Blanchflower-Oswald curve
Investment	Absent; capitalists' accumulated wealth is not reinvested	Retained profit is invested and accumulates a capital stock, the paper's central addition
Rationality	An independent variable of the study: graded endowments, from zero-intelligence to semi-rational	Held fixed; spending shares are frozen
Network	Its regularity is the other independent variable	Held symmetric
Inventories	Explicit; production adjusts through inventory accumulation and depletion	None; the firm cash buffer is an intra-period pass-through that returns to zero each period
Fiscal	A progressive tax funding a universal basic income	A flat tax funding an unemployment benefit, balanced-budget by construction (Section 8)
Method	Monte Carlo over a few scenarios	Global Morris/Sobol' sensitivity over sixteen parameters, with a committed claim registry, executable verifiers and continuous integration

The elasticity of substitution. Comparing economies that differ in σ is not straightforward: in the textbook CES, varying σ at fixed distribution and efficiency parameters also moves the implied factor shares, so any behavioral difference is unattributable. We therefore use the normalized CES (Klump and La Grandville, 2000; Klump and Saam, 2008; Klump et al., 2012), which fixes a common base point through which every σ-variant passes with the same factor shares. We follow Temple (2012) in reporting, rather than suppressing, the caveat that normalization is a comparison device and not an identification strategy: it makes σ-variants commensurable, it does not make σ a deep parameter. Section 6.3 accordingly reports the sensitivity of our central estimate to the choice of anchor.

Empirically, the literature places σ below unity: Chirinko (2008) weighs the evidence at 0.40–0.60; Chirinko and Mallick (2017) estimate ≈ 0.40; the meta-regression of Knoblach et al. (2020), covering 2,419 estimates from 77 studies, gives 0.45–0.87 and rejects the Cobb-Douglas hypothesis, a rejection consistent with earlier direct tests of the aggregate specification (Duffy and Papageorgiou, 2000; Antràs, 2004). A dissenting macro literature requires $\sigma > 1$ to explain the global decline of the labor share (Karabarbounis and Neiman, 2014). We treat this disagreement as a reason to sweep σ rather than choose it.

The wage curve. Wages are endogenized through the wage curve of Blanchflower and Oswald (1994), a level relation between the local wage and local unemployment with an elasticity near -0.10 (≈ -0.07

in the meta-analysis of Nijkamp and Poot, 2005). We use a level relation rather than a Phillips-type change relation deliberately: without trend inflation, a change relation produces perpetual wage drift at any $U \neq U^*$ and has no steady state, which would make the comparative-statics design of this paper impossible.

Distribution and effective demand. The question the model is built to answer is Kaleckian (Kalecki, 1971; Bhaduri and Marglin, 1990): whether a redistribution toward profits is expansionary or contractionary depends on whether the induced investment exceeds the lost consumption. Our contribution is not a new theory of that trade-off but a measurement of where its sign turns inside an explicit, stock-flow-consistent, disaggregated economy, and, more importantly, a measurement of how little of that sign survives when the rest of the parameter space is allowed to move.

3 The model

The economy is closed, single-good and fixed-price (the numéraire is 1). It is populated by $n_f = 10$ firms and $N = 100$ households, of which a share $\pi_c = 0.10$ are capitalists who own firms and receive dividends; the remainder supply labor. Time is discrete and one period is interpreted as one year (Section 4). Fig. 1 shows the agents and the money flows between them.

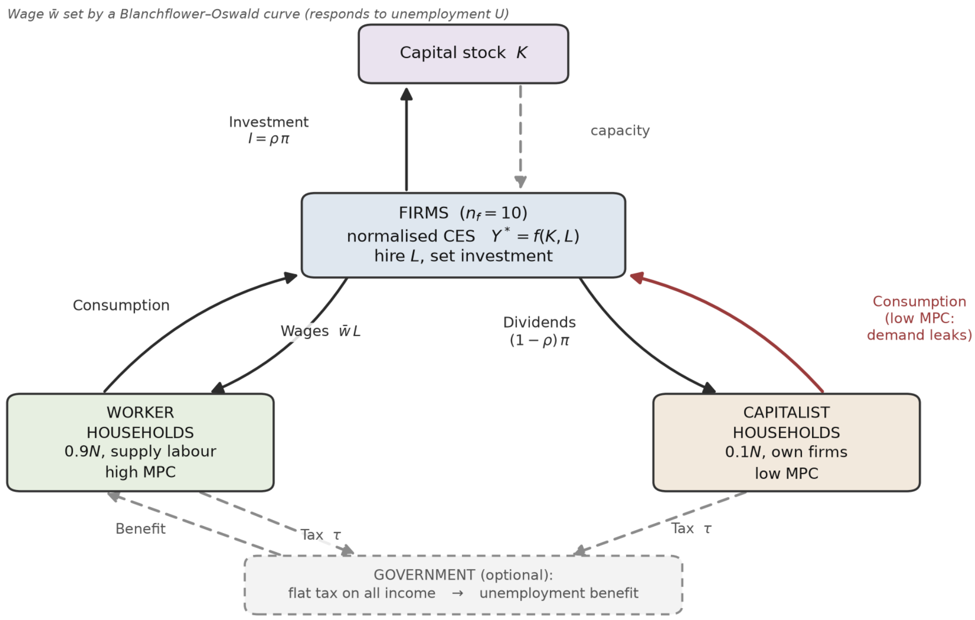


Figure 1: Structure of the model and its money flows. Firms produce with a normalized CES in capital and labor (Eq. (1)) and hire endogenously (Eq. (2)) at a wage set by a Blanchflower-Oswald curve; profit is split between retained earnings, invested as $I = \rho\,\pi$, which accumulates the capital stock, and dividends. Worker households supply labor and consume at a high marginal propensity to consume; capitalist households own the firms and consume at a low one, so income routed to them leaks from the circular flow (the channel drawn in red). The government block (dashed) is optional: a flat tax on all income funds a balanced-budget unemployment benefit (Section 8). The figure is a schematic and carries no measured quantity; it is produced by a committed generator, `scripts/make_fig_model.py`.

3.1 Production: a normalized CES

Each firm i has a potential output given by a normalized CES in capital and labor,

$$Y_i^* = Y_0\Big[\pi_0\big(K_i/K_0\big)^r + (1-\pi_0)\big(L_i/L_0\big)^r\Big]^{1/r}, \qquad r = \frac{\sigma-1}{\sigma}, \tag{1}$$

with realized output rationed by demand, $Y_i = \min\{D_i, Y_i^*\}$. Here (K_0, L_0) is the base point and π_0 the base-point capital share. Two properties matter.

Nesting Equation (1) nests Cobb-Douglas at $\sigma = 1$ and Leontief as $\sigma \to 0$. Because Y_0 is derived ($Y_0 = AK_0^{\pi_0}L_0^{1-\pi_0}$) rather than measured, the $\sigma = 1$ branch reproduces Cobb-Douglas exactly, for any base point.

Comparability (K_0, L_0, π_0) are measured once, at $\sigma = 1$ and $\rho = 0.40$, and then frozen across every experiment in the paper. Every σ-variant therefore passes through the same point with the same factor shares, so two economies differ only by σ. This is a declared modeling choice, not an estimate.

3.2 The labor market: three limits

Employment at firm i is the minimum of three distinct constraints:

$$L_i = \min\{L_i^{\text{demand}}, L_i^{\text{profitmax}}, N\}, \tag{2}$$

where L_i^{demand} is the labor needed to serve expected demand, $L_i^{\text{profitmax}}$ is the point at which the marginal product of labor equals the wage, and N is the aggregate workforce. Which limit binds identifies the regime: Keynesian (involuntary) unemployment, classical unemployment, or full employment respectively.

Model assumption 1 (Structural invariant: the workforce cap restores diminishing returns)**.** With a wage set outside the firm and an unbounded labor supply, $L_i^{\text{profitmax}} \propto K_i$, hence $Y_i^* \propto K_i$: the model becomes an AK economy with constant returns to capital, unbounded growth and no steady state. The cap $L \le N$ is therefore not a realism detail but the structural feature that restores diminishing returns, and it is asserted as such in the test suite.

Firms fire excess workers into an unemployed pool and fill vacancies by random matching; the discreteness of hiring and the randomness of matching are what give the model genuinely non-degenerate inter-seed variation.

3.3 Wages: a Blanchflower-Oswald wage curve

The wage is common to all firms and is set at the start of the period on the previous period's unemployment, which avoids a within-period simultaneity between w and U:

$$w_t = \max\left\{w_{\min}, \bar{w}\left(\frac{\max\{U_{t-1}, U_{\min}\}}{U_{\text{REF}}}\right)^{-\eta}\right\}. \tag{3}$$

The elasticity η is the parameter with empirical content; U_{REF} is the unemployment level at which $w = \bar{w}$, measured once at the anchor scenario and frozen; $U_{\min}$ and $w_{\min}$ are declared conventions (guards against $w \to \infty$ at full employment and against a deflationary spiral). At $\eta = 0$ the wage is constant at $\bar{w}$ and the model reduces bit-for-bit to its fixed-wage predecessor.

Profit is the residual, $\Pi_i = \text{sales}_i - w_t L_i$. There is no markup parameter: with a fixed price and a parametric wage, distribution is determined by w_t alone, and the wage share is a measured outcome, bounded above by the σ-dependent profit-max wage share.

3.4 Investment and internal financing

Firms finance investment entirely from retained earnings; there is no credit. Planned investment applies a retention ratio ρ to last period's profit, modulated by a capacity-utilization accelerator:

$$\phi_t = \max\{0,\ 1 + \beta(u_{t-1} - \bar{u})\}, \tag{4}$$

$$I_t^p = \text{clip}\left(\rho\, \Pi_{t-1}\, \phi_t,\ \underline{I},\ \Pi_t\right), \tag{5}$$

$$K_{t+1} = (1 - \delta)K_t + I_t. \tag{6}$$

The firm retains exactly what it invests and distributes the remainder as dividends. Utilization u is measured against profit-max capacity, not against realized output: with L chosen to serve expected demand, Y^* would otherwise track Y by construction and the accelerator would receive a dead signal.

The utilization entering Eq. (4) is itself a one-step adaptive expectation, $u_t^e = u_{t-1}^e + \lambda_u\,(u_{t-1} - u_{t-1}^e)$, and Eq. (4) is written for its default $\lambda_u = 1$, at which $u_t^e = u_{t-1}$ exactly. Every result outside Section 7.3 uses that default; Section 7.3 sweeps λ_u to test whether smoothing the signal changes anything, generalizing Eq. (4) rather than replacing it.

3.5 Households, government and stock-flow consistency

Households consume out of income and wealth, subject to a liquidity constraint,

$$C_h = \min\left\{ c_0 + c_1 y_h + \lambda a_h,\quad a_h + y_h \right\}, \tag{7}$$

with workers' c_1 above capitalists'. An optional minimal government (Section 8) levies a flat tax on accrued income to fund an equal transfer to the unemployed, indexed to the current wage, with the budget balanced every period by construction; a replacement rate of zero reproduces the no-government model bit-for-bit.

Model assumption 2 (Stock-flow consistency, and where it is tested)**.** The conserved quantity is $\sum_h(\text{wealth}_h + \text{income}_h) + \sum_i \text{buffer}_i$, and it is constant to within 10^{-9}. The firm cash account is an intra-period pass-through that returns to zero every period: an earlier design in which retained earnings accumulated as a stock between periods sequestered money and produced a spurious demand spiral.

Crucially, the invariant is tested across the parameter range a sweep will reach, not only at the default configuration. Section A.1 reports why: an ownership defect meant the invariant held at the default and nowhere else, destroying up to 97% of the money stock in 200 periods at $\pi_c = 0.05$, while the tests, run only at the default, stayed green.

3.6 Period sequence

The order of events is load-bearing and is stated explicitly (Algorithm 1). Employment is set before households form demand, because expected income depends on employment; investment settlement precedes household settlement; and the government step sits between them so that the tax falls on fully accrued income and the benefit arrives with the same one-period lag as a wage.

Algorithm 1 One period of the model

1: wage determination: set w_t from Eq. (3) on U_{t-1}
2: labor market: firms plan employment via Eq. (2); fire into pool; random matching fills vacancies
3: demand formation: households compute C_h (Eq. (7)); income is w_t if employed, else 0, plus dividends for capitalists
4: investment plans: firms apply Eq. (4)-Eq. (5)
5: demand registration: consumption + investment orders
6: production: $Y_i = \min\{D_i, Y_i^*\}$; goods market rations; utilization measured against profit-max capacity
7: firm accounting: wages paid, retention set aside, residual dividends declared
8: investment settlement: capital goods paid for, Eq. (6) applied, residual returned as dividends (buffer $\to 0$)
9: government: balanced-budget tax and transfer (skipped at $rr = 0$)
10: household settlement: income credited, goods paid for

4 Anchoring and calibration

A model of this kind can be calibrated in two quite different senses, and conflating them is how design targets come to be reported as results. We therefore assign every parameter to one of three anchoring tiers, and state the tier wherever the parameter is used.

(1) Anchored There exists an empirical quantity of which the parameter is an estimate, a citable source for it, and a comparator defined on the same perimeter as the model. σ, π_0, η, λ and the replacement rate are in this tier.

(2) Declared convention A value is needed, the literature offers a range rather than a point, and the choice within that range is ours. These are not estimates and are never described as such. δ is the consequential case.

(3) Non-anchorable by decision No empirical quantity corresponds to the parameter at all, so searching for a citation would be cosmetic. c_0 is the case, and it is handled by reporting both regimes throughout rather than by choosing.

Table 2 lists every parameter with its value and its tier. The tiers are not decoration: they record which numbers carry empirical content and which are modeling decisions that must not be presented as estimates.

Table 2: Parameters, values and anchoring tiers.

Parameter	Value	Anchoring
Anchored to a citable source		
σ (elasticity of substitution)	swept	0.40–0.60 (Chirinko, 2008); 0.45–0.87 and Cobb-Douglas rejected (Knoblach et al., 2020). Swept, never chosen.
π_0 (base-point capital share)	$1/3$	standard growth accounting (Gollin, 2002; Solow, 1957)
η (wage-curve elasticity)	swept, 0–0.15	$\approx$ 0.10 (Blanchflower and Oswald, 1994); $\approx$ 0.07 (Nijkamp and Poot, 2005)
λ (wealth effect)	0.05	$\approx$ 5 cents, 16-country mean (Slacalek, 2009); cf. 3–4 cents for housing wealth (Case et al., 2005)
ρ (retention ratio)	0.40	anchored through I/Y, not through payout ratios; see Section 11
rr (benefit replacement rate)	swept, 0–0.75	OECD net replacement rates $\approx$ 58% initial (OECD, 2024)
Declared conventions, not estimates		
δ (depreciation)	0.05	BEA-implied figure for this perimeter is $\approx$ 0.090; structures alone 2–3%. A defensible midpoint, not a measurement. See Section 9.3.
$U_{\min}$, $w_{\min}$	0.01, 0.45	guards; $w_{\min}$ never stably binds anywhere tested
max τ (tax cap)	0.6	guardrail; realized τ is a measured outcome
$\underline{I}$ (investment floor)	0.1	numerical guardrail
$\bar{u}$ (target utilization)	0.90	above the observed US average ($\approx$ 0.80)
Modeling choices, measured once, then frozen		
(K_0, L_0)	41.87, 7.395	CES normalization anchor (Klump and La Grandville, 2000); caveat per Temple (2012)
U_{REF}	0.26047	wage-curve normalization point. Not an estimate of the NAIRU.
$\bar{w}$	0.9	demoted to a normalization point once η exists
K_{init}	40.0	selects the basin (multiple equilibria); held fixed in every sweep
β (accelerator)	0.5	no canonical value exists; treated by sensitivity analysis
Declared non-anchorable		
c_0 (autonomous consumption)	$\{1.0, 2.0\}$	model units have no exchange rate with data (numéraire = 1). Its original justification was falsified; both regimes are reported everywhere.

Note. One model period = one year. This is a declared interpretive convention, chosen as the only one consistent with the already anchored parameters: δ is an annual rate and the Blanchflower-Oswald elasticity is estimated on annual data. It follows that the 2,000 steps of each run are a convergence device, not 2,000 years of simulated history, every result in this paper is comparative statics across steady states, never a time series to be overlaid on data.

4.1 The capital block, anchored flows first

The capital block is anchored flows first: ρ is anchored through the investment rate it produces rather than through corporate payout ratios, because ρ is what fixes I/Y here. The comparator is declared and held on both sides of the ratio, private non-residential fixed capital, since mixing perimeters is exactly how an earlier version of these notes derived a δ that appeared anchored and was not.

Against the BEA series for that perimeter, I/Y is 0.138–0.141 (U.S. Bureau of Economic Analysis, 2026). The model measured at $\rho = 0.40$ gives 0.158 in the anchor scenario and 0.182 in the headline scenario, above the anchor by two and four points. We quote the measured figure and never an analytical one: the relation $I/Y = \rho\alpha$ belongs to an earlier Cobb-Douglas core without a labor market and no longer holds once profit is a residual (Section A.4).

4.2 The anchored retention rate

Several statements in this paper turn on where the empirically anchored ρ sits, so it is derived here rather than assumed. The procedure is the one just stated, read backwards: ρ is anchored through I/Y, so the anchored ρ is the value at which measured I/Y enters the BEA band, by linear interpolation between the bracketing nodes of the committed sweep (Table 3).

Table 3: The anchored retention rate, derived from measured I/Y against the BEA band 0.138–0.141.

Scenario	I/Y at $\rho = 0.35$	Anchored ρ	Status
Anchor ($c_0 = 2.0$, $\sigma = 1$)	0.135	0.357–0.363	interpolated in support
Headline ($c_0 = 1.0$, $\sigma = 0.5$)	0.161	< 0.35	below the swept support

The two scenarios do not give the same number, and the difference cuts in the paper's favor rather than against it. In the anchor scenario the band is reached inside the support, at $\rho \approx 0.36$. In the headline scenario measured I/Y is already 0.161 at $\rho = 0.35$, the lowest value swept, so the anchored ρ lies below the support entirely. We decline to extrapolate it: a value obtained by extending the fit past the sweep would be a number about a region never visited.

What both scenarios support is an upper bound, and the bound is what the argument uses: the anchored ρ is at most 0.363. Wherever this paper says the anchored retention rate lies to the left of a turning point, the claim is that the turn lies above 0.363, which is checkable against any table of ρ^* and does not depend on resolving the anchored value exactly.

Remark 1. An earlier version of this project carried the anchored range as $\rho \in [0.35, 0.45]$, the lower end derived as above, the upper end not derived at all. The reported figures below use the derived bound. The committed sensitivity artifact still carries a side column computed under the wider band and therefore labels two low-σ cells as straddling the turn; Table 4 reports the corrected side, and the discrepancy is recorded rather than regenerated, since the artifact is evidence of what was run.

δ is the clearest case of tier (2). The depreciation rate implied by the BEA data for this perimeter is ≈ 0.090 (U.S. Bureau of Economic Analysis, 2025a), inflated by intellectual property products; structures alone sit at 2–3%, so the defensible empirical band spans roughly 0.02–0.09 and $\delta = 0.05$ is a midpoint chosen within it. It is not a measurement, and it must not be recalibrated "toward the center of the literature" late in a project: doing so would move every canonical number without buying any anchoring.

That decision has a structural consequence which Section 9.3 makes quantitative. The steady-state closure is $I/K = \delta + g$. This model has $g = 0$, so K/Y is not an anchor at all but a mechanical outcome: $K/Y = (I/Y)/\delta$, and measured I/K is 0.0500, exactly δ. Model K/Y is therefore 3.17–3.64 against a business-sector 1.23 (U.S. Bureau of Economic Analysis, 2025b), and no choice of ρ reconciles the two. The model cannot match business I/Y and K/Y simultaneously without trend growth, a declared structural limitation, not a calibration failure to be tuned away.

4.3 What is deliberately not anchored

c_0 is declared non-anchorable by decision: it is autonomous consumption in model units, and since the numéraire is 1 by construction there exists no empirical quantity of which it is an estimate. Searching for a citation would be cosmetic anchoring. Its sensitivity is instead handled by reporting both regimes $c_0 \in \{1.0, 2.0\}$ throughout, and by the global analysis of Section 9. This does not close the model's most visible quantitative gap, unemployment rates far above anything observed, which is a consequence of design decisions listed in Section 11, not of an unanchored parameter.

5 Experimental protocol

Steady states, not histories. Unless stated otherwise, every reported figure is a mean over 20 random seeds of the last 50 observations of a 2,000-step run. Runs are long because the transient must be exhausted, not because 2,000 periods of history are meaningful.

The sign frontier estimator. The quantity of interest is the sign of $\mathrm{d}Y/\mathrm{d}\rho$. We estimate an OLS slope of Y on ρ over the common viable support (the set of ρ values at which every configuration being compared survives), and define σ^* as the value of σ at which that slope crosses zero. Confidence intervals come from a percentile bootstrap over 2,000 resamples of the seeds. Restricting to a common support matters: comparing slopes estimated on different supports would confound the effect under study with differential survival.

Nesting checks as anti-drift sentinels. Every extension introduces a parameter whose null value must reproduce the previous model exactly: $\sigma = 1$ (Cobb-Douglas), $\eta = 0$ (fixed wage), $\lambda_e = 1$ (static expectations), $rr = 0$ (no government), $spread = 0$ (homogeneous firms). Each is implemented as an explicit branch and verified artifact-against-artifact on committed panels. The criterion under which those checks pass or fail is itself a measured object, and is reported in Section A.2: exact bit equality turned out not to be reproducible over time, and the replacement declares a tolerance on levels and zero tolerance on regime.

The reported numbers, and what may be quoted. Every figure in this paper is either measured from a committed driver script or labeled as a design target or convention. The rule is stated because it was adopted after an audit found previously recorded "results" that had never been measured, and because it has teeth: it is what required the withdrawals of Section A.4, and what forbids quoting an analytical identity in place of a measured ratio when the identity's derivation no longer applies. Its consequence is that results contradicting prior expectations are reported rather than recalibrated away, so that the negative findings are the paper's substance, not its residue.

Global sensitivity design. For Section 9, ρ is a treatment rather than a swept parameter: each design point is evaluated at several values of ρ under common random numbers (the same seeds at each, differenced or regressed per seed, then averaged). The reported analysis uses four values of ρ, and they contain the two of the superseded chord, so that both estimators are computed from the same runs and their difference cannot be a different sample. Without common random numbers the quantity of interest is a difference of two noisy numbers, and the sensitivity indices would decompose the variance of the seed draw rather than of the model. Uniform ranges over 16 parameters are a declared choice of ignorance. Screening uses the elementary-effects method of Morris (1991) with a pruning rule frozen in the source before any output existed; the survivors go to Sobol' estimation (Sobol', 2001; Saltelli et al., 2010) with bootstrap confidence intervals, computed with SALib (Herman and Usher, 2017). The number of seeds was fixed on evidence by a pilot: the noise-to-signal ratio measured 0.207, 0.170 and 0.111 at 3, 5 and 10 seeds, scaling as $1/\sqrt{n}$ as expected.

Implementation. The model is implemented in Python on the Mesa framework (Kazil et al., 2020). The automated suite, counted and described in Section A.6, covers stock-flow consistency (parametrized over the ownership range), per-seed determinism, labor accounting, every nesting limit, and directional regression pins. Each committed data file has a committed generator script; Section A.6 gives the details, and Section B collects the notation.

Data and code availability. All code, data, and the figure and table generators are in the repository at `https://github.com/MattiaSaramin/endogenous-investment-abm`, with citation metadata for this version in its `CITATION.cff`. Every table and figure is either produced by a committed driver script or explicitly labeled a design target or convention, and the correspondence between each output and its generator is recorded there (Section A.6). The headline numbers are held in a machine-readable registry, `paper_claims.yaml`, 49 claims in this version, each naming its committed artifact and the exact cell it is read from, which `verify_paper.py` checks against those artifacts, while `coherence.py` checks the same numbers for cross-document consistency; both are committed and runnable. One artifact is a declared exception: `ces_decomposition.csv` is archived rather than regenerable, and is cited by no table or figure.

6 Results I: the shape of $Y(\rho)$

The Kaleckian question asks for a sign, and a sign invites a single number. The first result of this paper is that the object being signed is a curve, and that stating its shape first makes every later result easier to read. We therefore begin with the response $Y(\rho)$ itself, and only then reduce it to the frontier σ^* that the literature, and the rest of this paper, uses as shorthand.

6.1 The response is U-shaped, and the turn moves with σ

$Y(\rho)$ is significantly curved. A quadratic fit places its turning point ρ^*, the argument minimum, inside the swept support in 19 of 22 (c_0, σ) cells, with $|t|$ up to 20.5. That single fact disqualifies any summary that compresses the response into one slope, because on a curve that turns inside the interval the sign of a slope depends on the interval chosen. Table 4 therefore reports the turn itself, together with the quantity that proves economically decisive: which side of it the anchored retention rate falls on, that rate being derived from the investment-rate anchor in Section 4.2 and bounded above by 0.363.

Table 4: Turning point of $Y(\rho)$ and the position of the anchored retention rate relative to it, at $\eta = 0$. The anchored ρ is at most 0.363, derived in Section 4.2; every turn below is above that bound.

σ	OLS slope	Chord	ρ^*	Anchored ρ lies
0.30	+41.9	+14.6	0.420	left of the turn
0.40	+26.8	+5.2	0.436	left of the turn
0.50	+17.8	−5.9	0.462	left of the turn
0.60	+5.6	−20.7	0.489	left of the turn
0.70	−4.8	−18.9	0.513	left of the turn
0.80	−11.7	−33.3	0.527	left of the turn
1.00	−29.6	−51.5	0.569	left of the turn
1.25	−45.3	−68.8	0.592	left of the turn
1.50	−61.1	−83.9	0.640	left of the turn

Two regularities run down that table. The turning point rises with the elasticity of substitution, from $\rho^* = 0.42$ at $\sigma = 0.30$ to 0.64 at $\sigma = 1.50$: the more readily capital substitutes for labor, the further retention must go before the supply channel overtakes the demand channel it displaces. And at every elasticity at which the turn is resolved, the whole range 0.30 to 1.50, the anchored retention rate lies to the left of the turn, on the falling branch. The margin is comfortable rather than marginal: the lowest turn in Table 4 is 0.420 against an anchored ρ of at most 0.363.

Remark 2. Retention depresses output while ρ remains below a σ-dependent threshold ρ^*, and expands it beyond; ρ^* rises with σ, from 0.42 at $\sigma = 0.3$ to 0.64 at $\sigma = 1.5$; and the anchored retention rate lies below ρ^* at every elasticity tested.

The economic content is a statement about the margin against the range. At the point where the data put ρ, the marginal effect of further retention on output is negative; over a wide enough increase in ρ it is positive. Both are properties of one curve, and neither is the label "wage-led" or "profit-led" applied to an economy. This formulation has the further virtue of not depending on which interval an analyst happens to pick, which is exactly the property a single-slope summary lacks, and which Section 9.5 shows to be worth 108 design points when the two are put side by side.

6.2 The Cobb-Douglas case as one point on that curve

At $\sigma = 1$ the anchored range lies well to the left of the turn, and a sweep over the canonical support reads unambiguously (Table 5).

Higher retention does exactly what the supply channel predicts to the capital stock, it rises from 256.5 to 453.2, and exactly what the demand channel predicts to everything else. Employment falls from 59.4 to 38.0, unemployment rises from 0.41 to 0.62, the wage share falls from 0.553 to 0.394, and output falls, from 96.7 to 86.8 (Fig. 2). This is the strongest possible reconnection to Teglio's leakage mechanism: investment itself becomes the leak.

Teglio (2025) anticipates this reading in the base paper's conclusions, which conjecture that reinvesting capitalists' accumulated wealth is unlikely to cure the shortfall in low-income households'

Table 5: Steady-state retention sweep at $\sigma = 1$, $c_0 = 1.0$. Means over 20 seeds, 2,000 steps, last 50 observations.

ρ	Y	K	Employment	Unemployment	Wage share	Utilization
0.35	96.7	256.5	59.4	0.406	0.553	0.687
0.40	91.1	281.0	51.9	0.481	0.513	0.591
0.50	87.3	346.6	43.9	0.561	0.452	0.459
0.60	86.2	414.0	39.3	0.607	0.411	0.379
0.65	86.8	453.2	38.0	0.620	0.394	0.349

consumption demand "at least in a model that does not envisage income growth". Ours is exactly that model, constant A and $g = 0$, so the conjecture is testable here, and the mechanism measured above is its quantitative counterpart on the one channel the model represents. The two propositions are adjacent, not identical, and the distinction is worth stating in the same breath: the conjecture concerns the wealth constraint on low-income households' spending, whereas what Table 5 measures is $\mathrm{d}Y/\mathrm{d}\rho$ at the anchored ρ; the negative margin there is the closest available counterpart to the conjecture on the accumulation channel, not a test of the household constraint itself. The conditioning clause it attaches, income growth "large enough", is exactly the no-growth limitation declared in Section 11, left here as a condition, not a result.

Note what Table 4 adds to that reading. The output column of Table 5 is not a verdict on the economy but the left branch of a U observed at one elasticity, and at the one value of σ the empirical literature explicitly rejects (Knoblach et al., 2020). Even here the turn is inside the support, at $\rho^* = 0.569$; the sweep falls because it is weighted toward the branch below it. Presented on its own, this table invites precisely the global label that Section 6.1 withholds.

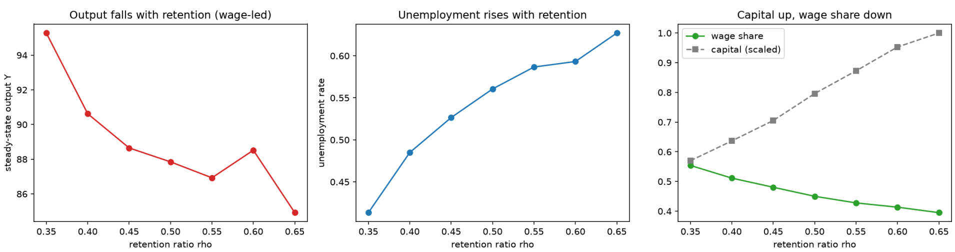


Figure 2: Retention sweep at $\sigma = 1$: output falls while capital rises. (Illustrative three-seed run; the canonical figures are those of Table 5.)

6.3 σ^*: the same structure read through a single slope

Reduced to one slope per cell, the structure of Table 4 presents as a frontier: the elasticity σ^* at which the estimated $\mathrm{d}Y/\mathrm{d}\rho$ crosses zero (Table 6). We retain the reduction, because it is the form in which the comparative results of Section 7 and Section 8 are legible, and because it is how the question is posed in the literature. But it is a reduction of the curve above, computed under the estimator declared in Section 5, an OLS slope over the common viable support, and it inherits that estimator's choices.

Two readings follow, and they pull in opposite directions.

The empirical range sits below the frontier With $\sigma \in [0.40, 0.60]$ and $\sigma^* = 0.654$, the empirically supported region has a positive whole-support slope: conditional on everything else being at its default,

Table 6: The sign frontier. OLS slope of the target on ρ over the common viable support; percentile bootstrap over 2,000 seed-resamples.

c_0	Target	σ^*	95% CI	Resamples with no sign change	$P(\sigma^* > 0.60)$
1.0	Y	0.654	[0.616, 0.691]	0%	99.8%
1.0	U	0.314	[0.300, 0.325]	0%	0%
2.0	Y	0.941	[0.918, 0.963]	0%	100%
2.0	U	0.450	[0.444, 0.456]	0%	0%

retention raises output across the swept range where the data put σ. Read against Table 4 this is the right branch of the U dominating the fit, not a contradiction of the negative margin at the anchored ρ, but the same curve summarized over a wider interval.

Unemployment has a different frontier $\sigma^*_U \approx 0.31$–0.45 lies well below σ^*_Y. In the band $\sigma \approx 0.3$–0.7, output and unemployment rise together, growth with technological unemployment, a configuration that neither "wage-led" nor "profit-led" describes. Figure 3 maps the sign over the whole (σ, ρ) grid.

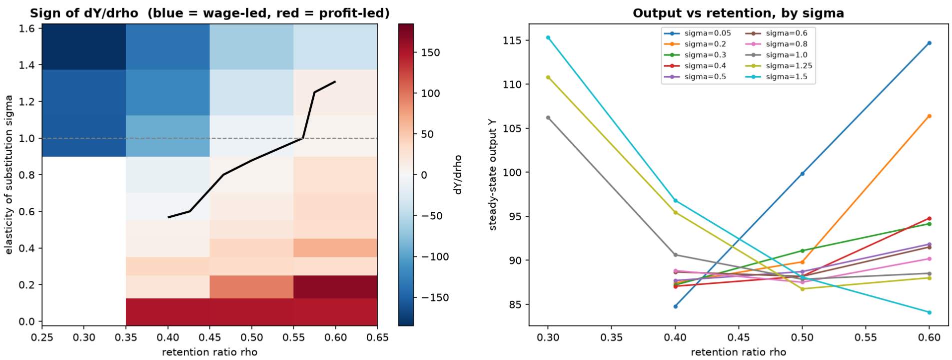


Figure 3: Sign of $\mathrm{d}Y/\mathrm{d}\rho$ over the (σ, ρ) grid.

6.4 What σ^* is conditional on

Reporting $\sigma^* = 0.654$ as a point estimate would overstate what has been measured. Beyond the curvature established in Section 6.1, two further conditioning dimensions move it materially. The choice of support: at $c_0 = 1.0$, σ^* ranges from 0.46 on $\rho \in [0.35, 0.55]$ to 0.94 on $\rho \in [0.45, 0.65]$. And the choice of normalization anchor: moving it from $\rho = 0.40$ to $\rho = 0.50$ shifts σ^* from 0.84 to 0.64, precisely the sensitivity Temple (2012) warns of, reported here rather than suppressed.

Both are consequences of the curvature, not independent fragilities. Where the labels "wage-led" and "profit-led" appear in what follows they are used as local descriptions of one side of the turn, under the estimator declared in Section 5; the convention is fixed here and not restated at each use.

7 Results II: stress tests and the wage channel

Each subsection below states a hypothesis that a critic might raise, turns on the corresponding mechanism, and reports what was measured. The unifying finding is not any one of them but what they share: a wage-driven feedback that erodes capital, discovered in the first test and unmoved by every instrument subsequently aimed at it, including the fiscal one of Section 8, which is reported separately because it also produces a result of its own.

7.1 Wage flexibility does not overturn the $\sigma = 1$ wage-led result, and it widens the profit-led band

The objection. The baseline holds the wage fixed, which suppresses an offsetting channel: unemployment should depress the wage, make labor cheaper, and slow the substitution toward capital. Turn that channel on and the wage-led result should weaken.

What was measured. The opposite of the objection happens, though not in the obvious direction (Table 7, Fig. 4). σ^* rises, so the $\sigma = 1$ wage-led result survives; $\sigma = 1$ stays above the frontier at every η, while the empirical band $[0.40, 0.60]$ is pushed deeper into the profit-led region. Wage flexibility does not rescue a wage-led reading of the empirically supported band; it enlarges the band in which retention raises output.

Table 7: $\sigma^*(\eta)$ on Y at $c_0 = 1.0$, estimated on the support viable at every η so that the columns are comparable. At $\eta = 0.10$ and $\eta = 0.15$ the slope $dY/d\rho$ crosses zero twice over the support (`n_crossings = 2` in `ces_b07_sigma_star.csv`); the entry is the first crossing, and is a single point only at the two smaller η, where the crossing is unique.

η	σ^*	95% CI	$P(\sigma^* > 0.60)$	mean U
0.00	0.654	[0.616, 0.691]	99.8%	0.528
0.05	0.666	[0.634, 0.692]	99.9%	0.543
0.10	0.725	[0.697, 0.745]	100%	0.565
0.15	0.740	[0.682, 0.793]	100%	0.579

σ^* rises with η, moving further above the empirical range, and mean unemployment rises with it ($0.53 \rightarrow 0.58$). The reason is that the counter-channel the objection invokes is not the only one the wage curve activates: a falling wage also shrinks the wage bill and hence demand. This is the Kaleckian paradox of costs, and in this model it dominates substitution. Wage flexibility does not auto-correct unemployment here; it makes it worse.

The price convention brackets this result. The result above holds the numéraire fixed at one, which makes the nominal wage the real wage, and that is a convention, not a measurement. A normal-cost probe relaxes it: with constant markup and productivity the price is $P_t = w_t/\bar{w}$, so a nominal wage cut is passed through to the price and the real wage w_t/P stays at $\bar{w}$. Under this complete, instantaneous price pass-through σ^* rises more steeply with η, not less (Table 8): the two conventions' intervals are cleanly disjoint at $\eta = 0.15$ and only marginally so at $\eta = 0.10$, separated there by about 0.001. The mechanism is a Pigou effect on real balances (lower prices raise real wealth, which raises demand) and it is stabilizing, so a reader who took complete pass-through for the truth would conclude that wage flexibility does auto-correct. Two cautions keep this a bracket rather than a verdict. The probe deliberately occupies the most stabilizing corner of the space (constant markup and productivity, no price stickiness), so it is an upper bound on how far pass-through can stabilize. And the off control is not itself resolved at twelve seeds: its σ^* is non-monotone in η, the endpoint falling below the $\eta = 0.10$

Figure 4: $\sigma^*(\eta)$: wage flexibility moves the frontier away from the empirical range.

value, and it carries two crossings at $\eta \geq 0.10$ where the frontier is not a single point. The honest reading is a bracket: at zero pass-through the wage curve does not auto-correct unemployment, at complete pass-through it does, and the model cannot say where between them the data sit. Section 10.4 returns to this as one of three senses in which the frontier is a local object.

Table 8: The price convention brackets the wage-flexibility result: $\sigma^*(\eta)$ on Y at $c_0 = 1.0$ with the numéraire fixed (off) and under complete instantaneous pass-through $P_t = w_t/\bar{w}$ (on), estimated on the support viable at every η. Vintage `ces_b21` (twelve seeds; an eleven-node σ grid), a different vintage from the `ces_b07` of Table 7 and not commensurable with it, the off column is the probe's own control, not Table 7. #cross counts the sign crossings of $\mathrm{d}Y/\mathrm{d}\rho$ over the support; the off control develops a second at $\eta \geq 0.10$, where σ^* is its first crossing.

η	pass-through	σ^*	95% CI	#cross
0.00	off ≡ on	0.648	[0.591, 0.709]	1
0.05	off	0.680	[0.657, 0.711]	1
	on	0.716	[0.671, 0.770]	1
0.10	off	0.726	[0.700, 0.750]	2
	on	0.806	[0.751, 0.838]	1
0.15	off	0.711	[0.658, 0.800]	2
	on	0.871	[0.843, 0.909]	1

A collapse mechanism, discovered here. In the secondary regime $c_0 = 2.0$, wage flexibility destabilizes the economy in the high-σ, low-ρ corner: at $\sigma = 1.25$, 43% of seeds collapse to $Y \to 0$, $U \to 1$ at $\eta = 0.15$, and σ^* becomes erratic or undefined. Tracing a collapsing trajectory identifies the mechanism, which turns out to govern much of the rest of this paper:

(1) the wage oscillates, rising above $\bar{w}$ when $U \to 0$ (the $U_{\min}$ guard drives w to ≈ 1.25) and falling below it when U is high; (2) as σ rises, $L^{\text{profitmax}}$ becomes more wage-sensitive, so the wage oscillation feeds a widening employment oscillation; and (3) each cycle, investment fails to cover depreciation, so capital erodes, and at low ρ the economy eventually dies at $U = 1$.

In the empirically relevant region $\sigma \approx 0.5$ the same curve leaves $w \approx \bar{w}$, produces no oscillation, and capital grows. The collapse is capital erosion at high σ, not a monotone wage spiral, and the subsistence floor $w_{\min}$ never stably binds, so this is genuine viability failure, not a floor artefact. We name this the wage $\rightarrow U \rightarrow$ capital-erosion channel; Section 10 returns to it.

The price probe separates the collapse. Step (2) of that mechanism turns on the firm reacting to the real wage: at a fixed numéraire the nominal wage is the real wage, so the wage oscillation is an oscillation in the firm's own marginal-cost signal. Under complete pass-through it is not, $w_t/P \equiv \bar{w}$, and step (2) does not start. Running $c_0 = 2.0$ with prices on does not, however, abolish the collapse. At $\eta = 0$, where $P \equiv 1$ by construction, the two conventions are identical (two cells collapse, one of them fully, under each). Away from there the incidence that climbed with η under the fixed numéraire, cells with any collapse running $2, 8, 12, 14$ across η, is flat with prices on, at $2, 3, 3, 2$; excluding the degenerate $\eta = 0$ cell, cells with any collapse fall from 34 to 8 and the fully collapsed from 16 to 3. The probe therefore separates the collapse into two components: a wage-driven part that pass-through removes, and a residual, already present at $\eta = 0$ and unrelated to wage flexibility, that it does not.

7.2 Adaptive expectations move neither the frontier nor the collapse

The hypothesis. A slower firm expectation should damp the oscillation just identified, and might also select a different steady state. We generalize the demand expectation from static to adaptive in the partial-adjustment form of Nerlove (1958), $Y_t^e = Y_{t-1}^e + \lambda_e(D_{t-1} - Y_{t-1}^e)$, which is the constant-gain case of the adaptive-learning literature (Evans and Honkapohja, 2001). We sweep λ_e rather than choosing it: that literature supplies the functional form, not a level, and no reliable point estimate exists for a firm's demand expectation in a model of this kind.

What was measured. The frontier is λ_e-invariant within confidence intervals (Table 9): every point lies inside its neighbours' intervals, and the empirical range stays below σ^* at every gain, so it is profit-led at every gain, while $\sigma = 1$ stays above and remains wage-led. What is robust to the expectation gain is the location of the frontier, and with it both readings on either side of it; so is its rise with η. There is no basin-selection effect.

Table 9: $\sigma^*(\eta; \lambda_e)$ on Y, $c_0 = 1.0$, across-configuration common support.

η	λ_e	σ^*	95% CI	$P(\sigma^* > 0.60)$
0.00	1.00	0.654	[0.616, 0.691]	99.8%
0.00	0.50	0.686	[0.637, 0.721]	99.9%
0.00	0.25	0.674	[0.639, 0.709]	99.9%
0.10	1.00	0.725	[0.697, 0.745]	100%
0.10	0.50	0.713	[0.667, 0.752]	100%
0.10	0.25	0.721	[0.684, 0.754]	100%

The stabilization half of the hypothesis is not confirmed. The collapse region is λ_e-invariant to within grid and seed noise (cells with any collapse at $\eta = 0.10$: 16, 15, 14 for $\lambda_e = 1, 0.5, 0.25$, with fully collapsed cells flat at 6; non-monotone at $\eta = 0.15$), and the reference collapsing cell dies at

$K = 0$, $U = 1$ at every gain. The interpretation is now sharp: damping the demand expectation cannot stabilize an instability that does not originate in demand.

7.3 The accelerator's signal, smoothed

The hypothesis. Section 7.2 smoothed the firm's demand expectation; the same question applies to the utilization signal the accelerator itself reads. Equation (4) reacts to realized utilization, and a hypothesis fixed before the runs held that letting firms react instead to the smoothed, adaptive $u_t^e = u_{t-1}^e + \lambda_u (u_{t-1} - u_{t-1}^e)$ would damp the accelerator and might move the sign result. We sweep the gain λ_u over $[0, 1]$; $\lambda_u = 1$ is the default of the main results, where $u_t^e = u_{t-1}$ exactly.

What was measured. The hypothesis is falsified on two levels. The mechanism fails at its root: the temporal standard deviation of the accelerator multiplier does not fall as λ_u decreases across $[0.25, 1.0]$, and reaches zero only at $\lambda_u = 0$. Utilization is highly persistent, and a constant-gain filter removes the variance of white noise, not that of a near-unit-root process, so smoothing an already-smooth signal changes little. The turning point ρ^*, the whole-support slope and the wage-led reading are correspondingly λ_u-invariant within inter-seed intervals across $[0.25, 1.0]$, at every β (Table 10). The finding is robust to how the accelerator's signal is formed.

Table 10: Anchored-margin resolution across accelerator strength β and smoothing gain λ_u, at the anchored headline configuration ($c_0 = 1.0$, empirical σ, $\eta = 0.10$), with ρ anchored at 0.363. Symbols: ↓ resolved on the falling branch (a negative anchored margin); ∼ unresolved, the anchor lies inside the ρ^* confidence interval (the margin vanishes); ∅ unresolved, the ρ^* interval is not estimable. Read from `ces_b17_margin`; `margin_resolved` is authoritative, so an absent interval is never read as resolved. The $\lambda_u = 0$ column switches the accelerator off: there $\rho^* = 0.358$ for every β, with the anchor inside the interval.

$\beta \setminus \lambda_u$	0.00	0.25	0.50	0.75	1.00
0.05	∼	∼	∼	∼	∼
0.15	∼	∅	∅	∼	∼
0.30	∼	↓	∼	∼	∼
0.50	∼	↓	↓	↓	↓
0.75	∼	↓	↓	↓	↓
1.00	∼	↓	↓	↓	↓

What this does and does not test. Two boundaries hold whenever the result is cited. First, the sweep is conditional on the anchored configuration: it varies λ_u and β at that point, not marginally over the full space, so it speaks to the anchored margin and not to the marginalized frontier of Section 9.1; the marginal counterpart was not run. Second, it holds $c_0 = 1.0$ and so never enters the $c_0 = 2.0$ collapse regime, where a demand-expectation gain showed its only effect (Section 7.2); that omission is a design gap of this experiment, met instead by the price probe of Section 7.1, whose grid includes $c_0 = 2.0$.

7.4 Firm heterogeneity: a cliff, not a gradient

All firms share one productivity. That is an assumption, and rather than implementing heterogeneity we measured what would happen if we did, using a single experimental dial that fans firm productivities out mean-preservingly, changing nothing else (no selection, no demand reallocation, no entry or exit).

The result is a cliff (Table 11). Below a threshold no firm dies; one grid step above it, every firm is dead, output is exactly zero, unemployment is exactly one, and aggregate capital decays geometrically to 3.5×10^{-34} by step 2,000.

Table 11: Aggregates against productivity dispersion (as U / Y).

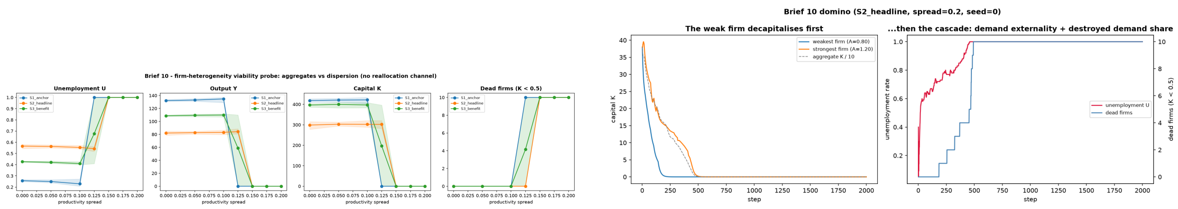
Dispersion	S1 (anchor)	S2 (headline)	S3 ($rr = 0.5$)	Dead firms (S2)
0.000	0.258 / 132.1	0.566 / 82.1	0.427 / 108.7	0
0.050	0.250 / 133.0	0.563 / 82.7	0.421 / 109.5	0
0.100	0.229 / 134.7	0.554 / 83.2	0.410 / 109.9	0
0.125	1.000 / 0.0	0.543 / 84.2	0.676 / 58.9	0
0.150	1.000 / 0.0	1.000 / 0.0	1.000 / 0.0	10
0.200	1.000 / 0.0	1.000 / 0.0	1.000 / 0.0	10

The domino. Tracing a collapsing run identifies the cascade: the low-productivity firm serves the same network demand with more labor, earns less profit, invests below δK and decapitalizes first (its capital falls from 38 to near zero by step 250). Its customers' spending shares remain pointed at it, so that demand is destroyed rather than reallocated, and its laid-off workers lose their income, a demand externality. The high-productivity firms follow: the strongest firm's capital reaches zero by step 500. This is precisely what the missing machinery would have prevented: with entry, exit and demand rerouting, that demand would have moved to a surviving firm.

The benefit again fails as a cushion. The hypothesis was that keeping income flowing to the laid-off would raise the viability threshold. It lowers it: at a dispersion of 0.125 the headline scenario has 0 of 20 seeds with any dead firm, while the same scenario with $rr = 0.5$ has 18 of 20, 7 of them fully collapsed. The verified mechanism is the same channel once more: the benefit lowers unemployment, the wage curve reads that and raises the wage, and the low-productivity firm, whose marginal product is scaled down by its own productivity, is the first pushed below $I = \delta K$.

What this licenses us to claim. Within-industry TFP dispersion is large: Syverson (2004) reports a 90/10 ratio near 2:1 in US manufacturing (see also Bartelsman and Doms, 2000). The collapse threshold here is a max/min productivity ratio of about 1.22–1.29. The units are not the same and no quantitative mapping is claimed, a linear fan half-width is not a 90/10 log-TFP ratio. The qualitative reading is all that is supported, and it suffices: empirically realistic firm heterogeneity lies well outside this model's viable range, because the model has no reallocation channel to absorb it. Building heterogeneous firms without building reallocation would produce a model that dies rather than a model with heterogeneous firms.

A collateral finding sharpens the mean-field question: even at zero dispersion firms do not stay identical, because consumption links are random. The capital share of the largest three firms starts at 0.30 and settles at 0.35–0.38. The defensible statement is that the firm side is quasi-representative in its aggregates, not in its cross-section. Figure 5 shows the cliff and the cascade together.



(a) The cliff.

(b) The domino.

Figure 5: Firm heterogeneity: aggregates are stable until they are not.

8 Results III: the sign region is conditional on fiscal institutions

The results so far describe an economy with no public sector, in which the unemployed earn nothing and therefore leak entirely out of the circular flow. That is a modeling choice, not a neutral baseline, and this section shows it carries the sign structure of Section 6 with it. We reinstate a minimal government: a flat tax on accrued income funds an equal transfer to the unemployed, indexed to the current wage, with the budget balanced every period. One economic parameter is added, the replacement rate rr, whose range is anchored to OECD net replacement rates (Table 2).

Dose-response: crowding-in. At the headline demand-constrained scenario the benefit lowers unemployment and raises both output and capital (Table 12).

Table 12: Fiscal dose-response at $c_0 = 1.0$, $\sigma = 0.5$, $\eta = 0.10$, $\rho = 0.40$.

rr	U	Y	K	Wage share	Realized tax	Cash-constrained
0.00	0.566	82.1	298.8	0.440	0.000	0.90
0.25	0.483	98.1	359.2	0.446	0.128	0.90
0.50	0.427	108.7	396.6	0.452	0.206	0.90
0.75	0.373	119.0	435.7	0.457	0.250	0.90

Capital rises from 299 to 436. In a demand-constrained regime redistribution is crowding-in, because more demand means more profit and hence more investment through $I = \rho\Pi$, the opposite of the crowding-out a loanable-funds account would predict, and a direct consequence of financing investment out of retained earnings rather than out of saving. The cash-constrained fraction stays at 0.90, all ninety workers, invariant to rr: the benefit never lifts a worker off the liquidity constraint, which is precisely why the balanced-budget multiplier keeps delivering across the whole dose.

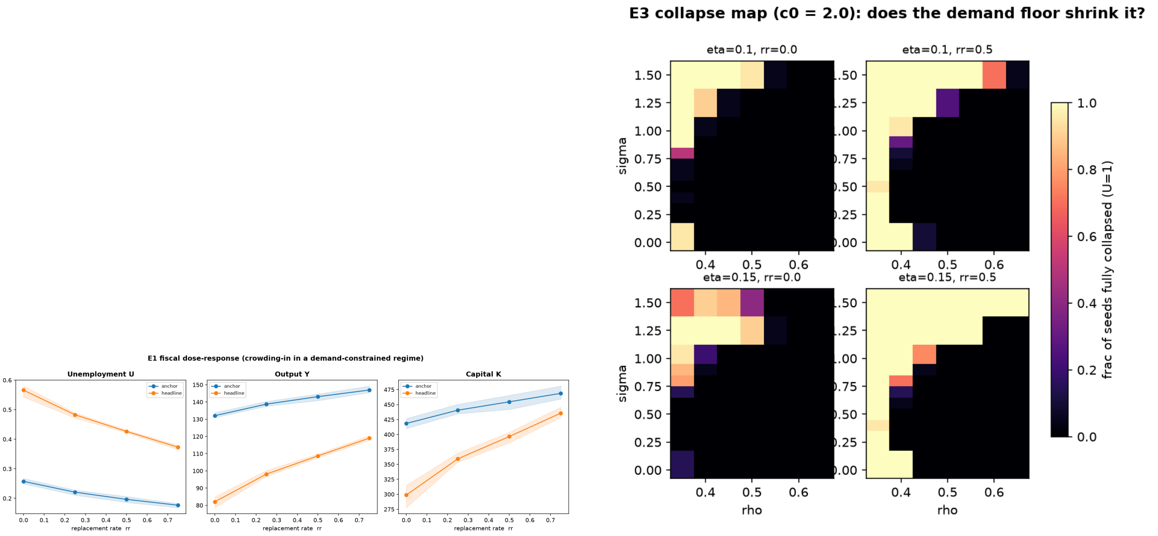
This direction is not new to the disaggregated Keynesian cross. One of Teglio's central results is that redistribution, there a progressive tax funding a universal income, lowers inequality and restores higher output; the same sign emerges here through different instruments, a flat tax funding an unemployment benefit with the budget balanced every period, and inside a model the base one does not have, with endogenous capital and an endogenous labor market. That a base-model result survives the addition of those two channels is a cross-validation rather than a fresh finding, and it is the survival, not the novelty, that makes it informative.

The benefit eliminates the wage-led region. At $rr = 0.5$, inside the empirically anchored band, σ^* becomes undefined: every estimated $dY/d\rho$ turns positive across the tested range (+38.7 at $\sigma = 1$;

+19.3 at $\sigma = 1.5$), pushing σ^* above 1.5. The demand floor makes retention expansionary at every elasticity tested. The unemployment frontier, by contrast, barely moves.

> *Remark* 3. The region in which retention depresses output is not a property of the technology alone. It exists because the unemployed leak out of the circular flow, and a balanced-budget benefit at a replacement rate inside the OECD band closes that leak and removes the region. The sign is conditional on fiscal institutions, and any statement of it that omits the institutional setting is incomplete.

The same instrument fails on the collapse. The falsifiable guess was that a demand floor would also shrink the collapse region of Section 7.1. It does the opposite: cells with any collapsed seed rise from 16 to 26 ($\eta = 0.10$) and 16 to 29 ($\eta = 0.15$), and the mean fraction of seeds at $U = 1$ roughly doubles. In the reference cell the economy dies at both $rr = 0$ and $rr = 0.5$, but at $rr = 0.5$ the tax is pinned at its cap in 100% of periods; the instrument saturates and the economy still dies. Three mechanisms explain it: where firms are collapsing the tax base is almost entirely wages, so taxing workers to pay workers is MPC-neutral; the benefit is indexed to w_t and is therefore procyclical, weakest exactly when it is most needed; and the demand it does inject amplifies the wage $\to U \to$ capital-erosion oscillation. That the same instrument can close the leak that produces the wage-led region and yet enlarge the collapse is the sharpest available statement of the boundary drawn in Section 10: it acts on demand, and only one of the two phenomena is a demand phenomenon.
Figure 6 shows the two faces side by side.



(a) Dose-response: U, Y, K against rr.

(b) Collapse map: the demand floor enlarges it.

Figure 6: The two faces of the balanced-budget benefit.

9 Results IV: global sensitivity

Every result so far is measured at a cell: a chosen σ, a chosen c_0, everything else at its default. We now ask what survives when all sixteen defensible parameters move at once. The quantity of interest is the one established in Section 6.1, an OLS slope of Y on ρ over four values of the retention ratio, evaluated under common random numbers, and not the two-point chord with which this analysis was first run. Why that distinction is worth a subsection of its own, rather than a footnote, is the subject of Section 9.5.

9.1 What marginalization removes is the frontier, not the sign

Remark 4. $P(\text{wage-led} \mid \text{viable}) = 0.026$ under the whole-support slope, on the empirically anchored band $\sigma \in [0.40, 0.60]$. Fraction viable = 0.480.

Over the empirically defensible space a negative whole-support slope is rare, and half of that space does not sustain an economy at all.

It matters what that number is about, because it is easy to read as the opposite of what it says. The quantity of interest here is the OLS slope over the whole ρ support, the global half of the statement in Section 6.1, the half that Section 6.3 already reports as positive in the empirical band. So $P = 0.026$ agrees with the conditional result rather than contradicting it: it says that a response which rises across the sweep is the norm once all sixteen parameters move. It says nothing about the turn ρ^*, and nothing about the sign of the margin at the anchored ρ, which is the paper's primary statement. Section 9.4 supplies that missing half.

What marginalization genuinely destroys is the frontier: the existence of a sign crossing at an elasticity inside the tested range. Conditionally $\sigma^* \approx 0.65$ sits in the middle of the swept σ; marginally the crossing is pushed to the edge of the range or out of it, and σ carries almost none of the variance. That is the claim this section destroys, and it is a narrower claim than "the headline". Table 13 places the headline beside the chord-based figures the same runs produce, since both estimators are computed from identical design points.

Table 13: The sensitivity headline under both estimators, computed on the same runs. $P(\text{wl})$ abbreviates $P(\text{wage-led} \mid \text{viable})$. The first row is the superseded chord-only analysis, retained for comparison.

Analysis	Fraction viable	$P(\text{wl})$ chord	$P(\text{wl})$ OLS
Chord-only, as first run	0.483	0.095	
Primary, $\sigma \in [0.40, 0.60]$	0.480	0.094	0.026
Wide, $\sigma \in [0.30, 1.00]$	0.468	0.202	0.098

The chord column reproduces the original analysis (0.094 against 0.095; 0.202 against 0.201), the residual being a changed survivor set at the screening stage rather than the estimator: under the identical keep rule and seed, `target_utilization` enters the surviving set and `benefit_replacement_rate` leaves it. A different quantity of interest genuinely selects different parameters, which is also why the earlier runs could not simply be recycled.

9.2 What actually governs the outcome

Two facts in Table 14 overturn the reading assembled to this point. First, $S_T \gg S_1$ throughout: the model is dominated by interactions, as one should expect of a wage $\to U \to$ capital channel that is interactive by construction. Second, and more uncomfortably, σ is close to irrelevant inside the empirical band,

Table 14: Sobol' indices. S_1 is the first-order index, S_T the total index; $N = 256$, bootstrap confidence intervals. Computed on the chord estimator (`ces_b13`); Section 9.2 reports the same decomposition on the OLS-repaired QoI (`ces_b14`), shown in Fig. 8.

	Viability		Slope	
Parameter	S_1	S_T	S_1	S_T
δ	0.718	1.002	0.331	0.966
π_0	0.116	0.290	0.001	0.562
c_0	0.069	0.086	−0.029	0.276
λ	0.038	0.086	0.039	0.264
σ	0.008	0.008	0.030	0.024

with a total index of 0.024 and a first-order index indistinguishable from zero. The depreciation rate and the base-point capital share do the work previously attributed to the elasticity of substitution.

The decomposition is robust to the estimator. Table 14 is computed on the chord, and repeating it on the repaired quantity barely moves it: on the slope, δ goes from 0.966 to 0.900, π_0 from 0.562 to 0.561, σ from 0.024 to 0.021; on viability, δ from 1.002 to 0.916. Which parameters generate the variance is robust to the choice of estimator; the level of the headline probability, as Table 13 shows, is not. These are different questions, and the answer to the first one stands under either quantity of interest.

9.3 The depreciation cliff

Table 15: Viability by depreciation rate. Chord estimator, screening vintage (`ces_b13`); bin edges are the sampled ones, rounded for display. See Table 13 for the OLS-repaired headline.

δ	Points	Fraction viable	P(wage-led | viable)
0.030–0.045	832	0.992	0.044
0.045–0.060	832	0.758	0.111
0.060–0.075	821	0.183	0.313
0.075–0.090	843	0.000	

Points and fractions above are read off the `ces_b13` binning; Section 4's $\delta \approx 0.090$ anchor and the `ces_b16` marginalized turning point (Table 16) are separate artifacts and must not be cross-cited as if interchangeable.

δ alone governs survival: its total index on viability is 1.00, and no other parameter approaches it. Table 15 is not a robustness note but a structural limitation, and it becomes one only when combined with the anchoring exercise of Section 4. The depreciation rate implied by the data for this model's perimeter, private non-residential fixed capital on both sides of the ratio, is ≈ 0.090. In that band, zero of 843 sampled points survive.

Remark 5 (Why this cannot be recalibrated away). The tempting reading is that the analysis confirms $\delta = 0.05$. It does the opposite, and the reason is the steady-state accounting identity $I/K = \delta + g$. The data satisfy it with $g \approx 0.022$; this model has $g = 0$, because technical change is out of scope. With $g = 0$, steady-state investment must cover all depreciation and nothing else. Raising δ toward the data without adding growth asks the capital block to finance a growing economy's depreciation out of a stationary economy's investment flow: capital erodes every period and the economy dies.

$\delta = 0.05$ is therefore not where the data put it. It is where $g = 0$ forces it to sit for the model to exist at all. The repair is a mechanism, trend productivity growth, not a calibration.

This is the same structural signature as the model's other declared failure of fit. Measured I/Y (0.158–0.182 at $\rho = 0.40$) sits above the BEA anchor of 0.138–0.141, and measured K/Y (3.17–3.64) sits far above the data's 1.23. The second follows mechanically from the first: with $g = 0$, $I = \delta K$ and hence $K/Y = (I/Y)/\delta$; measured I/K is 0.0500, exactly δ. The model cannot match business I/Y and K/Y simultaneously without growth. Seen from the level side, K/Y does not fit; seen from the viability side, the empirical δ kills the economy. One cause, two symptoms.

9.4 The turn, marginalized

The design evaluates every point at four values of ρ, so the turning point can be estimated per design point with the same quadratic fit Section 6.1 uses on the conditional grid, no further simulation, only a re-reading of the committed runs. This is the marginalized counterpart of the paper's first contribution, and it was missing from the analysis as originally reported.

Table 16: The turning point over the marginalized space, on the 1,596 viable design points. A turn is resolved only if the quadratic is convex, its curvature exceeds twice its own standard error, and the turn lands inside the swept support; the rest are reported undefined and are not extrapolated.

	Points	Fraction of viable
Convex fit	1,509	0.945
Curvature resolved at two standard errors	1,230	0.771
Turn inside the swept support	602	0.377
Resolved on all three	538	0.337
Where resolved: median $\rho^* = 0.414$ (p25 0.380, p75 0.451)		
Anchored ρ left of the turn: 477 of 538 = 0.887		

Two things follow. First, the U is frequently not observable in the marginalized space: only a third of viable points resolve a turn inside the support, and the median of all finite turns is 0.323, below the sweep. Where the turn is not observed we decline to extrapolate it, so the U-shaped statement of Section 6.1 remains a statement about the conditional cells and is not silently promoted.

Second, where the turn is resolved, the paper's primary claim survives marginalization and survives it comfortably: in 88.7% of those points the anchored ρ lies on the falling branch, so the marginal effect of retention at the empirically anchored point is negative. The global slope and the local margin point in opposite directions over most of the defensible space, which is not a contradiction but the definition of a U observed on one side of its turn.

Two declared limits. Four nodes resolve the turn coarsely. Refitting the conditional cells on the four SA nodes rather than their seven reproduces ρ^* to a median of 0.005 and a worst case of 0.014, and in none of the eleven cells does the coarser grid move the anchored ρ to the other side of the turn, so the resolution costs precision but not the decision. And ρ^* carries no variance decomposition: the Saltelli estimator needs a complete block of evaluations per base sample, and none of the 256 base samples has a resolved turn at every one of its thirteen evaluations. Imputing the missing turns would extrapolate exactly what the design cannot see, so the indices are omitted rather than reported weak.

9.5 The estimator is part of the result

The gap between the chord and OLS columns of Table 13 is not a rounding detail, and it is not diffuse. It was measured by a 2×2 experiment, method crossed with conditioning, under a verdict rule fixed before execution, whose full design, intervals and anchoring control are reported in Section A.3. The result of that experiment is that the quantity of interest was itself carrying part of the answer.

At identical parameters, changing only the estimator moves σ^* from 0.643 to 0.447 with non-overlapping intervals: for any σ between those values the chord is negative while the whole-support slope is positive. The two statistics genuinely disagree in sign on the same committed data. Conditioning moves the frontier further still, to 0.962, but it moves it away from the empirical band rather than across it, so under the declared rule the estimator is what the verdict turns on.

The difference is individually accounted for. Chord and OLS disagree in sign on 112 of the 1,596 viable design points; 110 of those are "chord negative, slope positive" and 2 are the reverse, for a net of 108. The headline gap is $(0.0940 - 0.0263) \times 1596 = 108$ points. The accounting closes exactly, and what distinguishes the disagreeing points is precisely the mechanism of Table 4: where the two estimators disagree the mean turning point is $\rho^* = 0.473$, inside the chord's $[0.35, 0.55]$ window; where they agree it is 0.820, outside the support, where $Y(\rho)$ is monotone and every estimator sees the same thing. A chord fixed while the response was assumed monotone kept returning answers about a curve it could not represent.

That the disagreement is concentrated where the turn falls inside the chord's window is also why it is not a defect of the model but of the summary: the two estimators are answering different questions about the same curve, and Table 16 shows how often that curve's turn is resolvable at all.

9.6 The wide-σ check

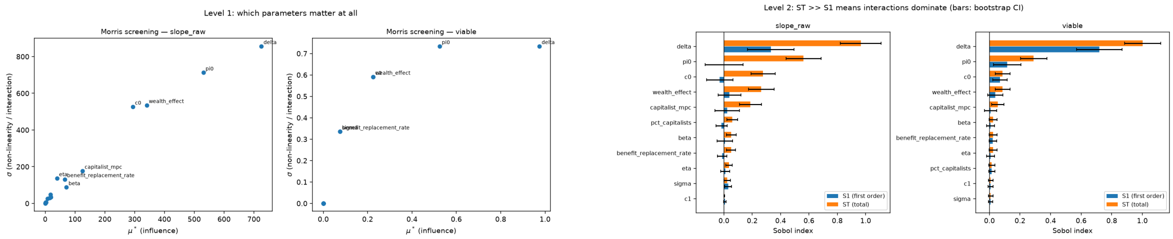
Widening the σ range to 0.30–1.00 leaves viability nearly unchanged but raises the wage-led fraction under both estimators, and the change is concentrated at high σ (Table 17).

Table 17: $P(\text{wage-led} \mid \text{viable})$ by σ bin under both estimators, on the same runs, $\sigma \in [0.30, 1.00]$.

σ bin	n viable	OLS	chord
0.301–0.475	191	0.000	0.016
0.475–0.649	197	0.025	0.127
0.649–0.823	201	0.154	0.299
0.823–0.998	190	0.211	0.363

Wage-led outcomes become more common as σ rises, under both estimators, the same direction as Section 6.3, which places the empirical band below σ^* and therefore on the side where retention raises output. The two analyses agree in direction; they differ in level, and the difference is the estimator. Note also that the probability never crosses 0.5 at any σ tested, topping out at 0.363: there is no threshold in the marginalized space at which wage-led outcomes become the majority.

Figure 7 shows the screening step and the variance decomposition that underlie Tables 14 and 15.



(a) Morris screening. **(b)** Sobol' indices with bootstrap CIs.

Figure 7: Screening and decomposition. Morris (left) screens seventeen parameters, the sixteen defensible parameters plus `max_tax`, which is screened here only, while the Sobol' decomposition (right) is over the sixteen-parameter defensible space. Both panels use the chord estimator (`ces_b13`); Fig. 8 is the OLS-repaired counterpart (`ces_b14`).

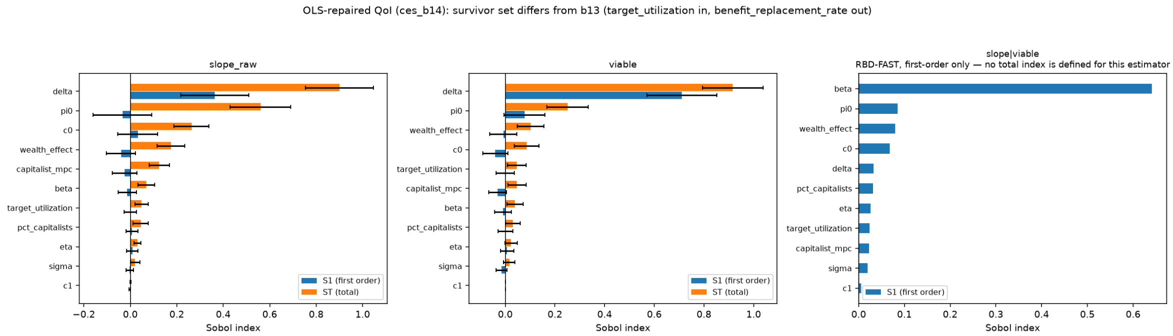


Figure 8: The same decomposition on the OLS-repaired QoI (`ces_b14`), the counterpart of Fig. 7's chord-based right panel. On the slope the total indices barely move from the chord values reported in Section 9.2: δ reads 0.900, π_0 reads 0.561, and σ reads 0.021; on viability δ reads 0.916. The survivor set differs from Fig. 7 (`target_utilization` enters, `benefit_replacement_rate` leaves), so the vertical axis is not identical. The third panel is the conditional quantity slope | viable under RBD-FAST, which defines only a first-order index, no total index is shown, and it is the panel in which the accelerator gain β dominates.

9.7 Two by-products

Kalecki, confirmed in levels. Within this model the relation $\Pi - I = C - W$ is a tautology and cannot settle causation. Sobol' sampling, however, varies the capitalists' propensity to consume independently of everything else, which makes the comparison interventional. Moving from the lowest to the highest quartile of capitalist MPC raises capitalist consumption by 10.83 and profit by 11.56, near one-for-one, with a correlation of +0.83 on levels. Capitalists do earn what they spend. The profit share falls slightly, because output rises faster; Kalecki's claim is about levels, and it is the levels that hold.

The accelerator produces the sign. Wage-led outcomes are overwhelmingly concentrated at high accelerator reactivity. In the lowest quartile of β ($[0.002, 0.251]$) only 0.6% of viable points are wage-led (2 of 353); in the highest ($[0.750, 0.999]$) the figure is 22.1%, and viability itself is higher (0.523 against 0.423). The wage-led sign is therefore substantially produced by investment reactivity rather than being an intrinsic property of the technology, and β, a parameter with no canonical empirical value (Table 2), governs both the sign and survival. This reverses the expectation that the instability would prove independent of investment reactivity, and it is an uncomfortable result: the headline depends on the one parameter in the model with the weakest empirical claim. A single-parameter decomposition makes the dependence quantitative: on the whole-support slope conditioned on viability, β ranks first of all parameters under RBD-FAST, with a first-order index of 0.370 on the chord estimator and 0.641 on the OLS-repaired quantity, so the ranking does not depend on the estimator repair. On the unconditioned slope the same β ranks only sixth by Morris μ^* (94.9), behind δ and π_0, which dominate there through viability rather than through the slope; the two readings are consistent, the conditional one isolating the accelerator once a viable economy is held fixed.

10 Discussion

10.1 One channel, three falsified hypotheses

The paper set out to measure a sign frontier and, in the process, found something more robust than the frontier itself. Three separate stabilization hypotheses were posed and each was falsified:

slower demand expectations would damp the collapse (Section 7.2), but they did not; a balanced-budget demand floor would shrink the collapse region (Section 8), but it enlarged it, with the fiscal instrument saturated at its cap; and that same benefit would cushion firm heterogeneity (Section 7.4), but it lowered the viability threshold.

All three fail through the same channel and for the same reason. The instability originates in the wage curve: unemployment moves the wage, the wage moves profit-maximizing employment (increasingly so as σ rises), employment moves unemployment, and each turn of the cycle leaves investment short of depreciation, so capital erodes. Demand-side instruments act on the wrong variable. Worse, in case (3) the benefit fails through a mechanism that is fully traceable and slightly perverse: by succeeding at lowering unemployment, it causes the wage curve to raise the wage, which pushes the weakest firm below its replacement investment.

There is a deeper reason the demand instruments miss, and it also says what would not: the loop turns because, at a fixed numéraire, the nominal wage is the real wage, so a wage cut is a real wage cut and no demand instrument can reach the variable that moves. A price convention can. Relaxing the numéraire to complete instantaneous pass-through switches the feedback off through a Pigou effect on real balances (Table 8), which is why the result is best read as bracketed by the price convention rather than settled. Section 10.4 places that bracket alongside two further senses in which the frontier turns out to be a local object.

The unifying boundary is worth stating plainly, because it is the most transferable result in the paper: demand instruments work where the binding constraint is demand, and not against wage-driven capital erosion.

10.2 What the global analysis does and does not overturn

The sensitivity analysis is destructive of the headline result of Section 6 and should be read as such: over the defensible parameter space, wage-led is the exception; σ is nearly inert marginally; and viability is governed by the depreciation rate, a parameter the calibration treats as a convention rather than an estimate (Section 4). Any version of this work that reported $\sigma^* = 0.654$ as the result would have been reporting a cell as if it were a space.

What it does not overturn is the conditional finding itself. A conditional frontier measured at defaults and a marginal effect averaged over fifteen other parameters are different objects, and Table 14 gives positive evidence that they should differ here. The honest summary is that the model has a robust local sign structure and a nearly inert global one.

10.3 The measurement instrument was part of the result

Section 9.5 reports the uncomfortable half of the reconciliation: a material part of the gap between the conditional and marginal readings was our own quantity of interest. A two-point chord and a whole-support slope are different functionals of a curved response, and on the same committed data they place σ^* at 0.447 and 0.643 with non-overlapping intervals. Re-running the global analysis on the repaired quantity moves $P(\text{wage-led} \mid \text{viable})$ from 0.095 to 0.026, and the whole of that movement is traceable to 108 individually identified design points, those where the turning point of the U falls inside the chord's window.

Two lessons generalize beyond this model. First, a summary statistic chosen before the shape of the response is known can encode a conclusion: the chord was fixed when $Y(\rho)$ was assumed monotone, and it kept returning answers about a curve it could not represent. Second, the fix was not to pick the better statistic but to stop compressing a curve into one number. That is why Section 6 leads with the turning point and the side the anchored range falls on, and treats the frontier as a declared reduction of it: of the three summaries, only the position relative to the turn is invariant to the analyst's choice of interval.

The episode has a plainer companion, recorded in Section A.4: an "inverted direction" between the conditional and marginal analyses, which this project reported as an open contradiction, did not exist in any measurement; it was an error in our own prose. The experiment designed to explain it was still worth running, because it found a real defect on its other axis. But the pairing is the lesson: a project's summary can contradict its own tables, and nothing except re-deriving each claim from the measurements will catch it.

10.4 The sign frontier is local; the anchored margin is not

Three findings, on three different designs, converge on the same qualification of the sign frontier, and are worth stating as one. Each shows the frontier moving, or dissolving, when a background choice is varied; none of them touches the result of Section 6.

First, the frontier is conditional on fiscal institutions: the balanced-budget benefit of Section 8 eliminates the wage-led region entirely at an OECD-band replacement rate. Second, it is nearly inert marginally: with the other parameters swept, the first-order sensitivity of the whole-support slope to σ, conditioned on viability, is 0.019 (Section 9), a frontier defined by a sign crossing in σ barely responds to σ once the rest of the space is allowed to move. Third, it is conditional on the price numéraire: the pass-through probe of Section 7.1 lifts σ^* upward, by roughly a grid node in σ at moderate wage flexibility and more at high η (Table 8).

What survives all three is not the position of the frontier but the anchored margin of the first result. Across fiscal settings, across the marginalized space, and across both price conventions, the anchored retention rate stays to the left of the turning point for $\sigma \geq 0.4$, so the marginal effect of retention at the anchored point keeps its sign. The frontier is a local, conditional object; the side the anchored range falls on is the robust one. Read this way the paper has a hierarchy of its own results, a fragile headline (σ^* as a number) resting on a durable one (the anchored margin of the U-shape), rather than a single frontier asked to carry more than a conditional cell can.

One qualification of that durability comes from the accelerator, and it cuts the other way. The anchored margin is robust to how the accelerator's signal is smoothed (Section 7.3), but it is load-bearing on how strong the accelerator is. At the anchored configuration the margin resolves as falling only for $\beta \geq 0.5$; below that it is unresolved, either because the anchor sits inside the turning-point interval or because that interval is not estimable (Table 10), and the default $\beta = 0.5$ sits on the boundary. Switching the accelerator off makes the point sharply: at $\lambda_u = 0$ the turning point is $\rho^* = 0.358$ for every β, with the anchor 0.363 inside its interval, so the margin does not invert, it vanishes; muting the accelerator the other way, at $\beta = 0.05$, gives the same outcome. The durable half of the hierarchy holds across elasticity, fiscal setting, price convention and expectation gain, but it rests on an accelerator reactive enough to place the turn to the right of the anchor, and that reactivity is the parameter (Section 9) with the weakest empirical claim in the model.

11 Limitations

We state these as declared limitations rather than as caveats, because several of them are consequences of design decisions taken deliberately.

No growth, and the capital block cannot fit the data without it With $g = 0$, the model cannot match business-sector I/Y and K/Y simultaneously, and at the empirically implied δ it is not viable at all (Section 9.3). This is a structural limitation with a known repair (trend productivity growth) that is out of scope here.

Unemployment is out of scale At the anchored retention rate and empirical σ the model runs at roughly 50% ($c_0 = 1.0$) or 31% ($c_0 = 2.0$) unemployment. No tested value of c_0 brings it into a plausible band. This is a structural tension of the labor-market design, reported as an open question rather than calibrated away. The Section 8 benefit is a genuine lever here: it lowers headline unemployment from 0.57 to 0.37, but it is a policy instrument, not a fix for the tension.

The wage share is too low The wage share sits below its empirical range, at 0.35–0.61 across viable cells against an empirical 0.60–0.68, touching it only at high σ and low ρ.

Multiple equilibria Initial capital selects the basin, so it is held fixed across every experiment and reported. Cells that collapse are an outcome, not an error.

No reallocation, so no genuine heterogeneity Section 7.4 closes the firm-heterogeneity question by measuring that the model cannot absorb empirically realistic dispersion; entry, exit and demand rerouting are declared future work.

No credit, no endogenous prices, and the fixed numéraire is an unanchored convention Internal financing is a design choice to be defended as such, not an omission; with the numéraire fixed at one, the implicit markup is already an outcome rather than a parameter. Endogenizing it would require markup data of the kind assembled by De Loecker et al. (2020), and is declared future work. The numéraire is not neutral for the wage-flexibility result: Section 10.4 shows it brackets that result, a normal-cost probe with complete instantaneous pass-through switching the wage $\rightarrow\ U\ \rightarrow$ capital-erosion channel off through a Pigou effect, and the model cannot say which convention the data sit near. A closer variant that lets unit cost move endogenously was declined deliberately, not overlooked: it pins the wage share by construction, and so would settle the question by assumption rather than measure it.

Sampling limits in the global analysis Three seeds leave roughly 4.3% of the quantity's variance as seed noise, which lands in the residual, depressing first-order indices and inflating apparent interaction. With $N = 256$ over eleven parameters, several first-order indices are indistinguishable from zero.

The turning point is often unresolved marginally The U-shaped statement of Section 6.1 is a statement about the conditional cells. In the marginalized space ρ^* falls inside the swept support in only 37.7% of viable points (the in-support fraction of Table 16, wider than the share at which the turn is fully resolved there), and below it in 58.7%. Where the turn is not observed we decline to extrapolate it, so the primary formulation is not available everywhere the sensitivity analysis samples.

Two estimates of the marginalized frontier, not one The bridging experiment sweeps all fifteen non-σ parameters and puts $\sigma^* \approx 0.96$; the wide Sobol' design holds five of them at midpoints and does not cross zero below $\sigma = 1$. They agree in message: the marginalized frontier sits at or above $\sigma \approx 1$, but they are different designs and we do not merge them into a single number.

The headline rests on the least anchored parameter Section 9 finds the wage-led sign to be produced by the accelerator reactivity β, the parameter with no canonical empirical value (Table 2), and Section 10.4 sharpens this into a dependence: at the anchored point the margin resolves only for $\beta \geq 0.5$, with the default on that edge (Table 10). The result is robust to how the accelerator's signal is formed but load-bearing on how strong the accelerator is, and the strength has no data behind it. The durable half of the paper's hierarchy is itself anchored to a modeling choice.

12 Conclusion

We have built a stock-flow-consistent agent-based economy in which investment is endogenous, capital accumulates, labor is hired endogenously, and the wage responds to unemployment, and in which, consequently, the Kaleckian question of whether retention expands or depresses output is an outcome rather than an assumption.

The first result is that the outcome has a shape rather than a label. The response $Y(\rho)$ is U-shaped, and the statement that survives every choice of estimator is about the turn rather than the sign: retention depresses output while ρ lies below a σ-dependent threshold ρ^* and expands it beyond, ρ^* rises with σ, and the anchored retention rate lies below ρ^* at every elasticity tested. Summarized through one slope that structure appears as a sign frontier at $\sigma^* \approx 0.65$, which rises when wages are made flexible, the paradox of costs dominating substitution, and is invariant to expectation dynamics. We use "wage-led" and "profit-led" in that reduced sense throughout, as local descriptions of one side of a turn, and not as a verdict on the economy.

The second is that the region in which retention depresses output is not a property of the technology alone. A balanced-budget unemployment benefit at a replacement rate inside the OECD band eliminates it, making retention expansionary at every elasticity tested, by closing the leak through which the unemployed drain the circular flow. A sign reported without its institutional setting is incomplete, and the same instrument that closes the leak is the one that, applied to the collapse, makes it worse.

The third is the finding to carry forward, and it was not an anticipated one. It is the wage → unemployment → capital-erosion channel, which the wage curve produced without being sought, killed three consecutive stabilization hypotheses, and proved immune to every demand-side instrument applied to it. Demand policy works where demand binds. Where the wage bill drives an oscillation that erodes the capital stock, more demand is not a stabilizer, and in the sharpest case measured here, a saturated fiscal instrument financed by taxing workers to pay workers simply amplifies the cycle that kills the economy.

Two boundaries are placed on all three. Marginalized over the full defensible parameter space the conditional structure largely dissolves: σ explains about 2% of the variance and survival is governed almost entirely by the depreciation rate, at whose empirically implied value the model does not exist at all, which, read with its inability to match I/Y and K/Y jointly, identifies one structural culprit in a capital block without trend growth. And the sign itself proved partly an artefact of how we measured it: a two-point chord across a U-shaped response is not the whole-support slope that defined the frontier, and replacing it moves the headline probability from 0.095 to 0.026 while leaving the variance decomposition intact. Neither boundary was anticipated when the quantities were chosen, which is the argument for choosing them after the shape of the response is known rather than before.

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A Validation and reproducibility

This appendix collects the checks the paper’s results rest on, the two occasions on which those checks failed, and the claims withdrawn as a result.

Remark 6 (Two principles, each learned from a measured failure). An invariant tested at a point holds at that point. A property verified at one configuration carries no information about any other, and invariants must be parametrized over the range a sweep will reach before the sweep rather than after.

A check that reports success without inspecting anything is worse than no check, because it converts “not measured” into “measured and sound”. The absence of a check is visible; a check that cannot fail is not.

Both are stated as principles because both were measured. The stock-flow invariant of Section A.1 held only at the default configuration while the test suite stayed green throughout, and two of this document's own build detectors reported clean counts on a log that contained plenty (Section A.6).

A.1 An invariant that held only at the default

Firm ownership was assigned by cycling over households: capitalist i owned firm $i \bmod n_f$. That is a bijection only at the default configuration (100 households $\times$ 0.10 = 10 capitalists, 10 firms). Off the default it broke in both directions (Table 18).

Table 18: The ownership defect, measured at seed 0 over 200 steps.

π_c	Capitalists	Firms with an owner	Money t=0 $\rightarrow$ t=200	$\sum$ net worth vs K
0.05	5	5/10	400.00 $\rightarrow$ 11.34	0.53×
0.08	8	8/10	400.00 $\rightarrow$ 46.15	0.84×
0.10	10	10/10	400.00 $\rightarrow$ 400.00	1.05×
0.20	20	10/10	400.00 $\rightarrow$ 400.00	2.10×

Below the default, an ownerless firm's dividends and residual cash were paid out inside a guard that tested for an owner, so they simply vanished, a direct violation of stock-flow consistency. Above it, the assignment overwrote itself, leaving capitalists holding stale references whose capital was still summed into net worth, inflating measured wealth inequality by a factor of two.

What makes the defect worth a subsection is when it was found: immediately before the global sensitivity analysis, which sweeps exactly the parameter that exposed it. Undetected, Section 9 would have reported variance decompositions for a model that leaks money. The fix cycles over firms, so every firm has exactly one owner for any number of capitalists; at the default the assignment is unchanged and a slice of the committed panels reproduces exactly, so no previously reported result moves.

A.2 The reproducibility criterion, and its two declared limits

The nesting checks of Section 5 are verified artifact-against-artifact on committed panels. The original criterion, deviation exactly 0.0, proved not reproducible over time. Code that had recorded 7/7 exact reproductions later deviated by 1 unit in the last place (ULP) on the same cells; eight hypotheses were excluded by measurement and the cause remains unidentified. The magnitude across 160 cells and 24 metrics is at most 2.1 ULP, does not amplify, and produces zero regime flips, so no economic conclusion moves. Re-running the reference slice under the replacement gives 7/7 with a worst significant drift of 0.00 ULP; it reproduces exactly today, where the earlier measurement found 2.1 ULP on the same code. That the drift is intermittent is the strongest argument for retiring exact equality: a criterion that passes or fails by the day carries no information about what changed in the model.

The replacement has two limbs that cannot compensate for each other: a declared tolerance on levels (8 ULP with an absolute floor, roughly four times the measured envelope), and a check on regime indicators (viability, which constraint binds, the sign of every resolvable metric) at tolerance exactly zero, since that is where drift would become scientific rather than numerical. Two limits are declared in the source. A pure ULP tolerance is unusable near zero, a metric legitimately resting at zero registering thousands of ULP on a gap of 10^{-16}; hence the absolute floor. And it must not be applied to differenced or fitted quantities (a chord, a slope, a curvature) because catastrophic cancellation makes their relative error arbitrary even from stable inputs: the sensitivity analysis's slope deviates by 3410 ULP while its own inputs deviate by 4. Those are checked by sign instead.

A.3 The bridging experiment in full

Section 9.5 reports the verdict; this is the design that produced it. The experiment is a 2×2: method (chord versus OLS) crossed with conditioning (all other parameters fixed at their defaults versus marginalized over the sensitivity ranges), on a shared grid of runs. The verdict rule was fixed before execution: a sign flip across the method axis at fixed parameters implicates the estimator; a flip across the conditioning axis at constant method implicates conditioning; both flipping means both contribute and the weights must be quantified; neither flipping is a finding to be reported, not forced. The fixed-parameter arm cost no simulation, the committed panels already carrying the canonical grid at twenty seeds, so only the marginalized arm required new runs. Table 19 reports the design.

Table 19: Bridging experiment, complete. The fixed arm holds every parameter at the configuration in which Section 6.3 identified σ^* ($c_0 = 1.0$, $\eta = 0$); the marginalized arm draws the other fifteen parameters from the Section 9 ranges on the same σ grid. All rows use the four-point ρ support shared by both arms, so the comparison is not confounded by the support.

Cell	Parameters	Method	σ^*	95% CI
1	fixed	chord	0.447	[0.395, 0.501]
2	fixed	OLS	0.643	[0.596, 0.701]
3	marginalized	chord	0.938	[0.573, 0.976]
4	marginalized	OLS	0.962	[0.723, 0.985]

The mandatory anchoring control passes to fourteen significant figures: cell 2 evaluated on the full committed support returns $\sigma^* = 0.6540142777288407$ against the published 0.654 with interval $[0.616, 0.691]$. The bridge is therefore measuring the same object as Section 6.3, and any difference across the table is the treatment.

What the rule does not capture, reported anyway. The rule asks only whether the empirical band changes sides, and by that test conditioning does not fire: cells 2 and 4 both place the band below σ^*. But conditioning moves σ^* further than the estimator does, and with non-overlapping intervals. Reporting only the verdict would let a threshold stand in for the measurement; both displacements push the same way, and the empirically supported band is profit-led more robustly than Section 6.3 alone claimed.

A.4 Withdrawn claims

The declared rule is that every number entering a table must be measured from a committed driver or labeled a design target (Section 5). Applying it retrospectively required withdrawing three claims, recorded here rather than silently corrected.

A contradiction that did not exist An earlier summary reported an inverted sign direction between the conditional and marginal analyses and left it open as a contradiction. There was none: "the empirical band sits below σ^* and is therefore not wage-led" and "below $\sigma \approx 0.65$ wage-led outcomes are rare" are the same statement. The error was in our prose, not in any measurement. The bridging experiment above was designed to explain the inversion and survived the discovery that half of what it set out to explain was a reading error; its method axis still measured a real disagreement, and is reported as it was fixed, not as it would have been written knowing the answer.

An inverted label A summary of the adaptive-expectations result claimed "the wage-led result is robust to the expectation gain", citing as evidence that the empirical band stays below σ^*, which is the profit-led side. The measured λ_e-invariance is unaffected; the label attached to it was wrong, and Section 7.2 states it correctly.

A stale identity The relation $I/Y = \rho\alpha$ was carried forward from an earlier Cobb-Douglas core with no labor market and used to reason about anchoring after it had ceased to hold; measured I/Y is cited in its place throughout (Section 4). A related error mixed comparators, business-sector I/Y against whole-economy K/Y, and produced a δ that appeared anchored. Both are withdrawn.

A.5 A gate that fired open

The λ_u sweep of Section 7.3 was run behind a decision rule frozen before the runs: it would count as movement any cell whose anchored margin changed state as the gain fell. The rule returned open, on six triggers. Decomposed after the fact, the six are four degenerate cells at $\lambda_u = 0$, where the accelerator is switched off and the margin vanishes for every β; two near-anchor cells at $\beta = 0.05$, where the anchor sits on the turning point and the state is seed noise; and zero cells showing a gradient in the smoothing gain itself. The substantive reading is that nothing moved with λ_u on $[0.25, 1.0]$, and the question was closed on that substance, over an open gate. The rule's mechanical verdict is recorded here unaltered, because the defect was in the gate's design rather than in the finding: it admitted the degenerate $\lambda_u = 0$ control into a test meant to detect movement in the gain, and a control built to show no accelerator is not evidence that the gain matters. This is reported for the same reason as the withdrawn claims above, not because anything here was withdrawn.

A.6 Reproducibility

Every table in this paper comes from a committed output file with a committed generator script; the correspondence is recorded in the repository. Runs pin BLAS thread counts before importing numerical libraries, because thread-dependent reduction order would otherwise break per-seed determinism, and the execution environment, library versions and the frozen sampling seed, is recorded alongside the sensitivity-analysis outputs.

The suite comprises 569 automated tests. Beyond conventional unit coverage they assert the model's economic invariants directly: stock-flow consistency parametrized over the ownership range rather than at the default alone (Section A.1), per-seed determinism, exact nesting at every limiting parameter value, the workforce cap that restores diminishing returns (Section 3), and directional regression pins on basin selection. Fifteen cover the repaired quantity of interest and the criterion of Section A.2, pinning the reason for each declared constant rather than the constant itself, including, as an executable statement of the defect this paper repairs, that a chord and a whole-support slope can carry opposite signs on the same U-shaped response.

Two detectors that reported a reassuring zero. The second measured instance of this appendix's opening principle concerns the present document rather than the model. The job that compiles the paper counts overfull and underfull boxes in the TEX log. Both counters matched the literal string `\hbox` through a layer of shell quoting that consumed the backslash, so both matched nothing and both reported zero on logs containing many: a green check that had inspected nothing, read as evidence of typographic cleanliness. The counts quoted for this document were produced after the detectors were repaired and verified against a synthetic log with known boxes.

B Notation

Symbol	Meaning
σ, r	elasticity of substitution; CES exponent $r = (\sigma - 1)/\sigma$
σ^*	the elasticity at which $\mathrm{d}Y/\mathrm{d}\rho$ changes sign
ρ, ρ^*	retention ratio; the turning point of $Y(\rho)$
π_0, (K_0, L_0), Y_0	base-point capital share, base point, derived scale
η, $\bar{w}$, U_{REF}	wage-curve elasticity, normalization wage, normalization unemployment
λ_e	expectation gain (adaptive demand expectations)
rr, τ	benefit replacement rate; realized tax rate
δ, β, $\bar{u}$	depreciation, accelerator sensitivity, target utilization
c_0, c_1, λ	autonomous consumption, propensity to consume income, wealth effect
π_c	share of households that are capitalists
S_1, S_T	first-order and total Sobol' sensitivity indices